

A Phenomenological Correspondence Between Catuskoti Logic and Binary Logic

Masaomi Chiba

2025-12-31

Abstract

This paper presents a phenomenological specification of Catuskoti (fourfold) logic and binary logic within a unified framework. The goal is not foundational justification or truth evaluation, but a descriptive correspondence based on observation, projection, fixed points, and complexity-based distance measures. Binary logic is treated as a restricted observational projection of the same underlying structure described fully by Catuskoti logic.

1 Scope and Assumptions

The present work adopts the following constraints:

- No claims of correctness, proof, or logical superiority are made.
- All descriptions are phenomenological and observational.
- Truth is treated as a geometric or informational relation, not a boolean value.
- Dynamics, temporal evolution, and proof theory are explicitly excluded.

2 Shared Underlying Structure

Both Catuskoti logic and binary logic are assumed to operate on a common underlying space of propositions or descriptions. Differences between the two arise solely from:

- The choice of observational projection.
- The definition of distance used to identify “truth.”

A distinguished point Φ is defined in this space and referred to as the *middle-way fixed point*.

3 Correspondence Table

Observation Aspect	Catuskoti Logic	Binary Logic
Underlying space	Proposition space	Same space
Observation projection	Fourfold projection	Two-valued projection
Accessible phases	Is, Not-Is, Both, Neither	Is / Not-Is only
Fixed point	Middle-way fixed point Φ	Same fixed point Φ
Fixed point behavior	Stable	Repulsive
Truth definition	Minimum Kolmogorov distance to Φ	Maximum distance from infinity
Truth behavior	Convergence to Φ	Divergence toward extremes
Undecidability	Internal phase	Explicit indeterminacy
Observation outside projection	None	Intermediate phases

4 Incompleteness as an Observational Effect

Incompleteness phenomena are interpreted as consequences of projection choice, not as failures of the underlying structure.

4.1 Godel-Type Phenomena

In binary logic, certain propositions become undecidable due to the restriction to two-valued observation. In Catuskoti logic, the same propositions occupy intermediate observational phases and remain representable.

4.2 Chaitin-Type Phenomena

When truth is defined via extremal distance criteria, complexity bounds appear as external limitations. Under minimum-distance evaluation, these bounds are treated as observable phases rather than obstructions.

5 Interpretive Summary

- Catuskoti logic provides a complete observational decomposition of the proposition space relative to the fixed point Φ .
- Binary logic corresponds to a coarse projection of the same space.
- Both systems share the same fixed point, but assign it opposite stability roles.
- Incompleteness arises from projection, not from the space itself.

6 Conclusion

This specification demonstrates that Catuskoti logic and binary logic may be treated as isomorphic descriptions of a single underlying structure, differing only in observational resolution and distance-based truth criteria. No dynamical, proof-theoretic, or metaphysical claims are required for this correspondence.